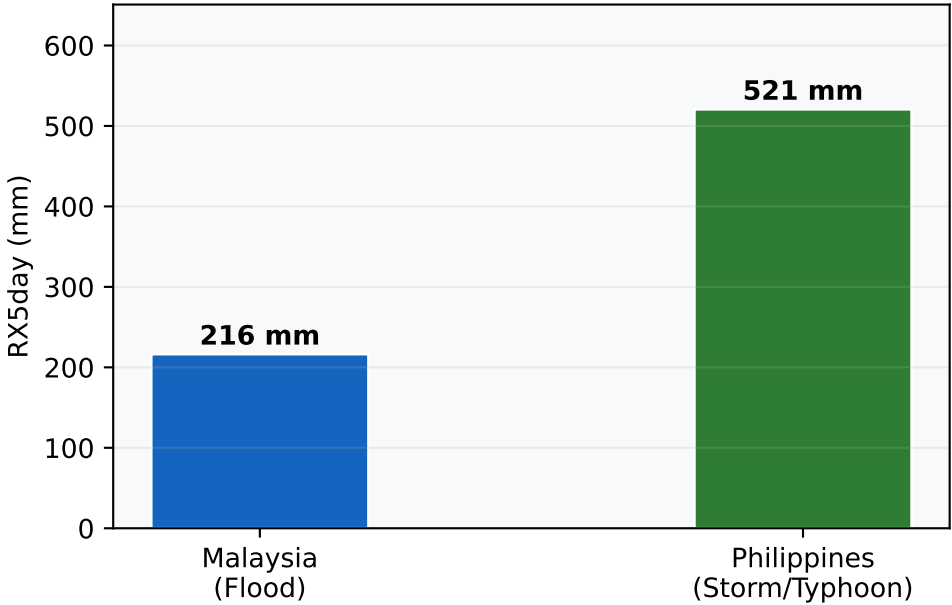


# Hannover Re SEA Treaty Book — Physical & Transition Risk · Three-Action Framework

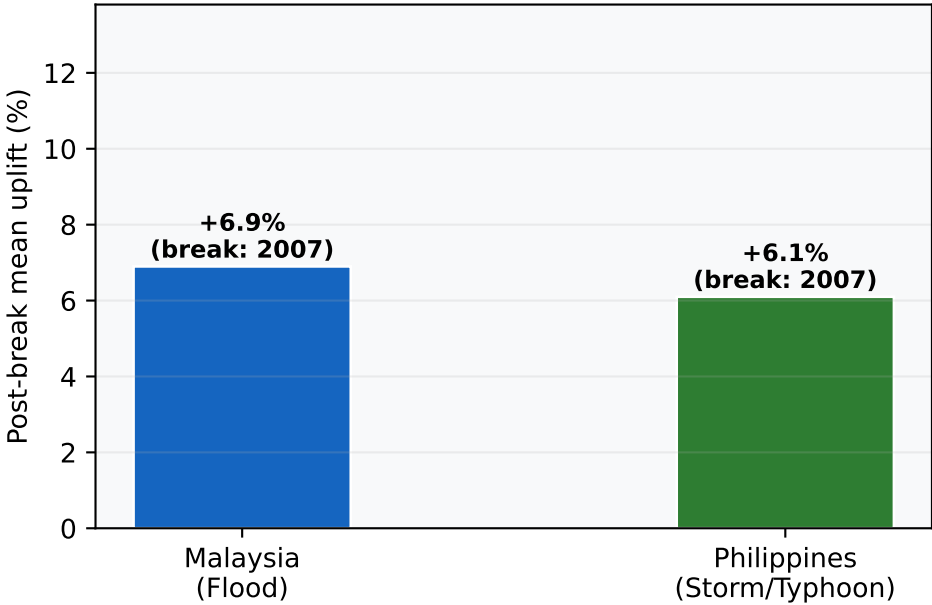
R5 Executive Summary | Malaysia & Philippines | Pre-2007 Hazard Transition Risk Quantification

Physical gap: 100-yr flood is structurally underpriced



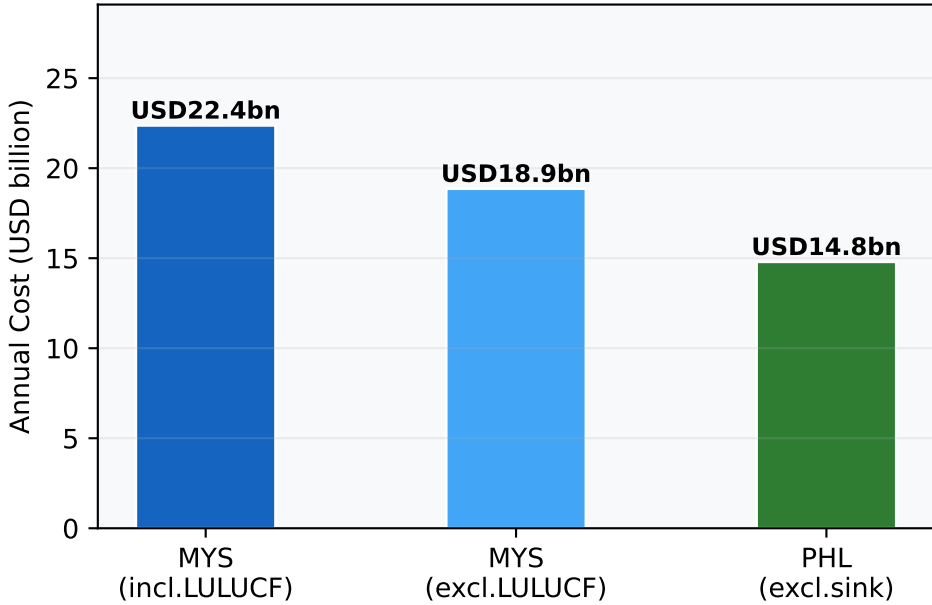
CHIRPS RX5day (WMO ETCCDI) · GEV MLE fit · Weibull family ( $\xi < 0$ )

Regime shift: post-2007 hazard baseline +6-7% unreserved



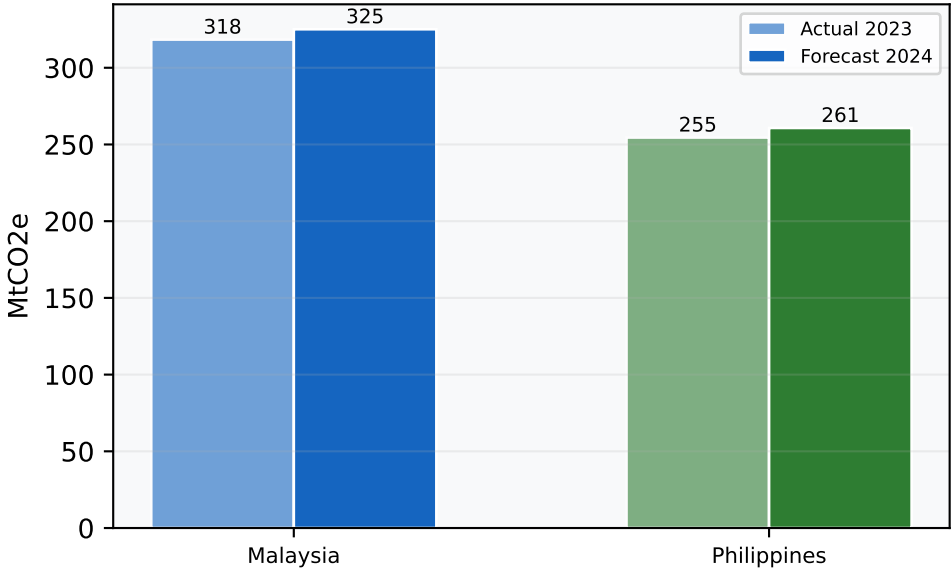
ruptures PELT · RBF kernel · RX5day pre vs post-break mean

Transition gap: MYS faces USD 22.4bn/yr compliance cost



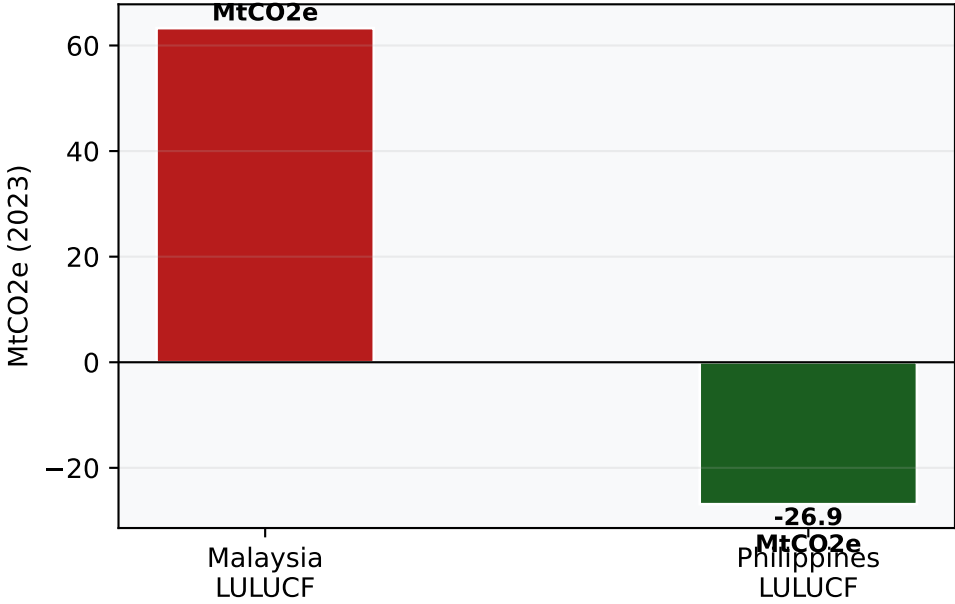
NGFS GCAM 6.0 · USD55.58/tCO<sub>2</sub>e · Climate Watch 2023

GHG still rising: ARIMA confirms no peak by 2024



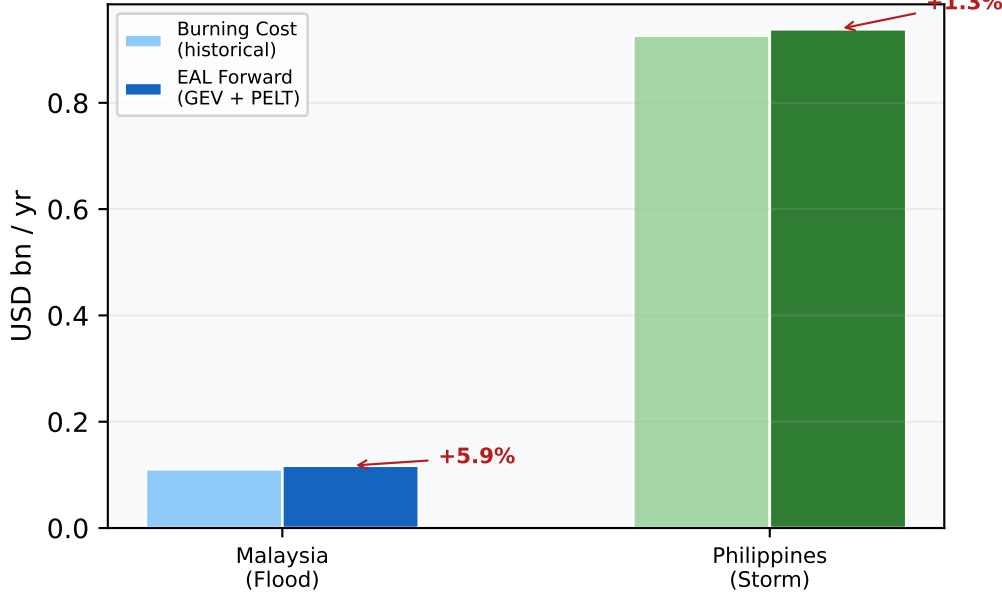
ARIMA(1, 1, 1) MYS MAPE=1.7% · ARIMA(2, 1, 0) PHL MAPE=3.6%

Uniform SEA factor is wrong: MYS emits, PHL absorbs LULUCF



Climate Watch 2023 · Drives asymmetric transition liability

Premium inadequate: GEV EAL exceeds burning cost in both markets



GEV analytical mean · PELT uplift · EM-DAT (1991–2024) burning cost

## (1) Recalibrate Physical EAL (Return Period Compression)

### DATA

MYS 100-yr flood: 216 mm | EAL gap +5.9%  
PHL 100-yr storm: 521 mm | EAL gap +1.3%  
PELT regime break: 2007 (both markets, synchronized)

### ACTIONS

- MYS: +6–8% flood loading on XL layers (EAL gap +5.9%; +1–2pp for n=34 CI margin)
- PHL: +1–2% floor loading ONLY (CI [326–985mm] is 3× wide — no point-estimate)
- Replace burning cost with GEV EAL (treaties > USD 50M)

CHIRPS v2.0 · GEV MLE · ruptures PELT · EM-DAT 1991–2024

## (2) Apply Country-Specific Transition Risk Surcharges

### DATA

MYS: USD 22.4bn/yr  
LULUCF emitter: +63 MtCO<sub>2</sub>e (palm oil / deforestation)  
PHL: USD 14.8bn/yr  
LULUCF net sink: -27 MtCO<sub>2</sub>e (reforestation credit)

### ACTIONS

- MYS: 3–5% surcharge (BNM CCPT C3/C4 cedants)  
Add Climate Warranty Clause at renewal
- PHL: 1–2% surcharge (LULUCF sink → Art. 6.2 ITMO credit partially offsets)
- Re-evaluate at every NGFS Phase update

NGFS GCAM 6.0 · BNM CCPT 2021 · BSP Circ.1085 · UNFCCC Art.6

## (3) Pass-Through Governance & ENSO Forward Protocol

### DATA

PT rate 1%→5% = 5× treaty exposure variance:  
SEA: USD 0.37–1.86bn/yr  
HRe (10% alloc): USD 3.0M–14.9M/yr  
ENSO: ONI r = -0.016, p = 0.93 — absent at annual scale

### ACTIONS

- Apply 5% PT to C3/C4 if carbon > USD 111/t  
Review annually at each NGFS Phase release
- ENSO is a null finding, not a pricing driver  
Use NOAA La Niña 12-mo outlook as TCFD facultative audit trigger — NOT a pricing variable

NGFS GCAM 6.0 · NOAA ONI 1950–2026 · BNM CCPT tiers